THE NOVA RESEARCH CURRICULUM

Chemistry

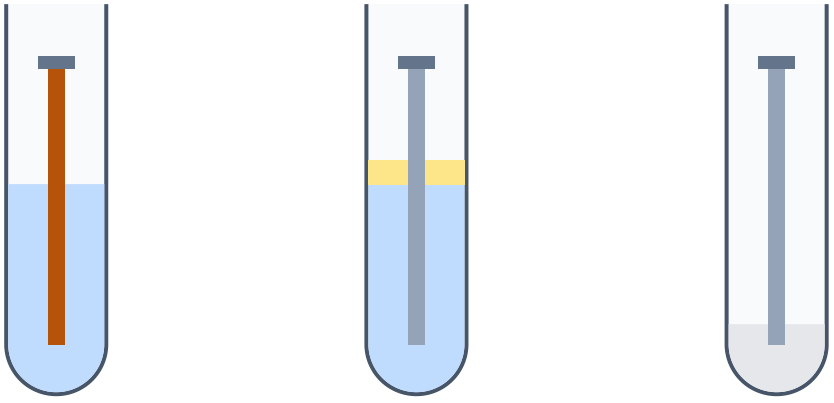
Grade 7 · Term 2

Student edition

Outcome: CH-7.2 physical and chemical changes, rusting and combustion.

Lesson notes

- **Changes (Weeks 1–4):** physical change: no new substance, usually reversible (melting ice, dissolving sugar, cutting paper). Chemical change: new substance formed, usually not reversible; signs are gas, colour change, heat or light, a precipitate (burning wood, cooking an egg, milk turning sour). Home lab: baking soda and vinegar.
- **Rusting (Weeks 5–8):** iron + oxygen + water → iron oxide (rust). Practical: three nails in dry air, boiled water under oil, and tap water over a week. Prevention: paint, oil or grease, galvanising (zinc coating), stainless steel.



Tube A: air + water
rusts

Tube B: boiled water + oil
no rust

Tube C: dry air (drying agent)
no rust

Rusting needs both oxygen and water.

- **Combustion (Weeks 9–12):** fuel + oxygen → carbon dioxide + water + heat and light. Fire triangle: fuel, heat, oxygen — remove one to put a fire out. Fire safety at home, gas-leak safety, and using a fire blanket. Teacher demonstration only for burning.

Workbook pages

Page	Task
1–2	Sort 16 changes into physical and chemical with reasons.
3	Rusting practical report.
4	Fire triangle and home fire-safety plan.

Question bank

#	Outcome	Question	Your answer
1	CH-7.2	Which is a chemical change?	melting ice / dissolving salt / burning paper / cutting wood
2	CH-7.2	Melting chocolate is a physical change. True or false.	True / False
3	CH-7.2	Rusting needs iron, oxygen and ...	salt / water / sunlight / carbon dioxide
4	CH-7.2	Galvanising coats iron with ...	copper / gold / zinc / plastic
5	CH-7.2	The three sides of the fire triangle are fuel, heat and ...	water / oxygen / nitrogen / smoke
6	CH-7.2	A fire blanket works by ...	adding fuel / cooling only / cutting off oxygen / adding heat
7	CH-7.2	Give two signs that a chemical change has happened.	_____
8	CH-7.2	Burning fuel gives off carbon dioxide and water. True or false.	True / False